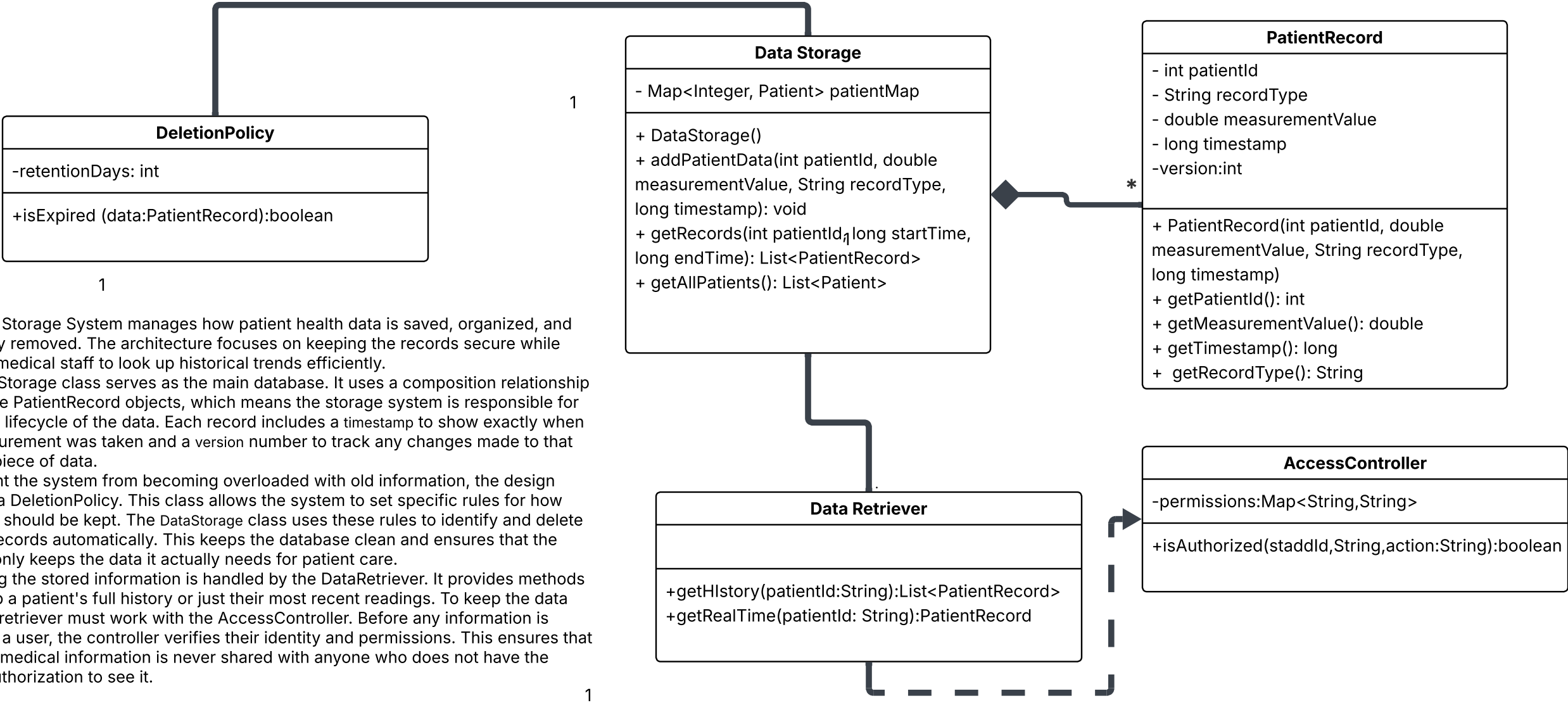




The Data Storage System manages how patient health data is saved, organized, and eventually removed. The architecture focuses on keeping the records secure while allowing medical staff to look up historical trends efficiently.

The DataStorage class serves as the main database. It uses a composition relationship to manage PatientRecord objects, which means the storage system is responsible for the entire lifecycle of the data. Each record includes a timestamp to show exactly when the measurement was taken and a version number to track any changes made to that specific piece of data.

To prevent the system from becoming overloaded with old information, the design includes a DeletionPolicy. This class allows the system to set specific rules for how long data should be kept. The DataStorage class uses these rules to identify and delete expired records automatically. This keeps the database clean and ensures that the hospital only keeps the data it actually needs for patient care.

Accessing the stored information is handled by the DataRetriever. It provides methods to look up a patient's full history or just their most recent readings. To keep the data safe, the retriever must work with the AccessController. Before any information is shown to a user, the controller verifies their identity and permissions. This ensures that sensitive medical information is never shared with anyone who does not have the proper authorization to see it.